

The Perceptron

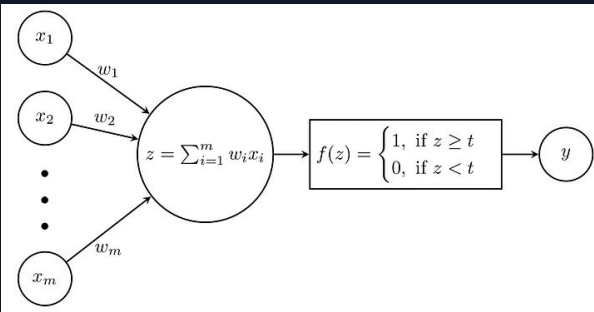
The elemental predecessor of neural networks

Prof. Didi Ramsaran Chin

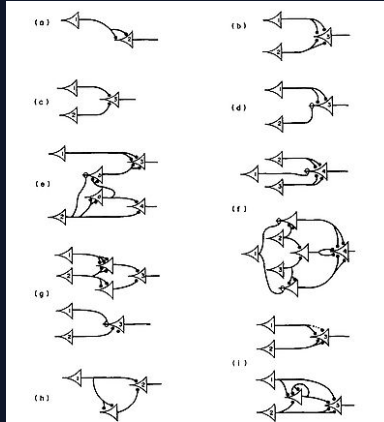
INFO-02020

September 2026

Timeline of the perceptron

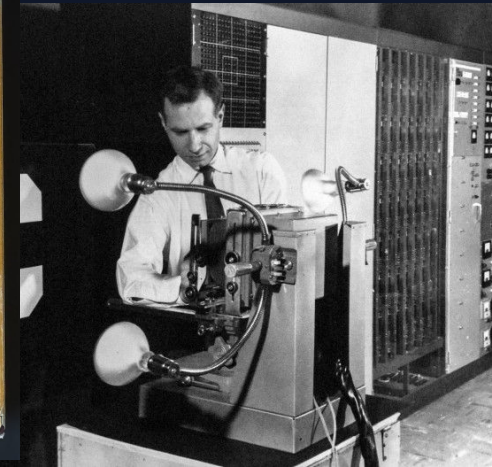
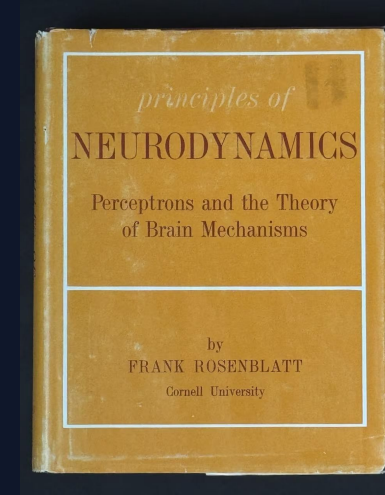


[Source](#)



1939–1952 - the squid

Hodgkin and Huxley combined biology and physics to create a circuit-like model (and its equations) of action potential



[Source: Wikimedia Commons](#)

1943 - Cambridge

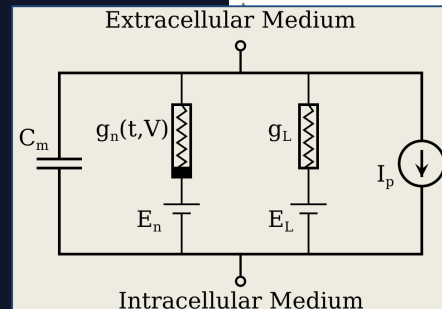
McCulloch and Pitts established the relation between neurons and boolean logic.

A QUANTITATIVE DESCRIPTION OF MEMBRANE CURRENT AND ITS APPLICATION TO CONDUCTION AND EXCITATION IN NERVE

By A. L. HODGKIN AND A. F. HUXLEY

From the Physiological Laboratory, University of Cambridge

(Received 10 March 1952)

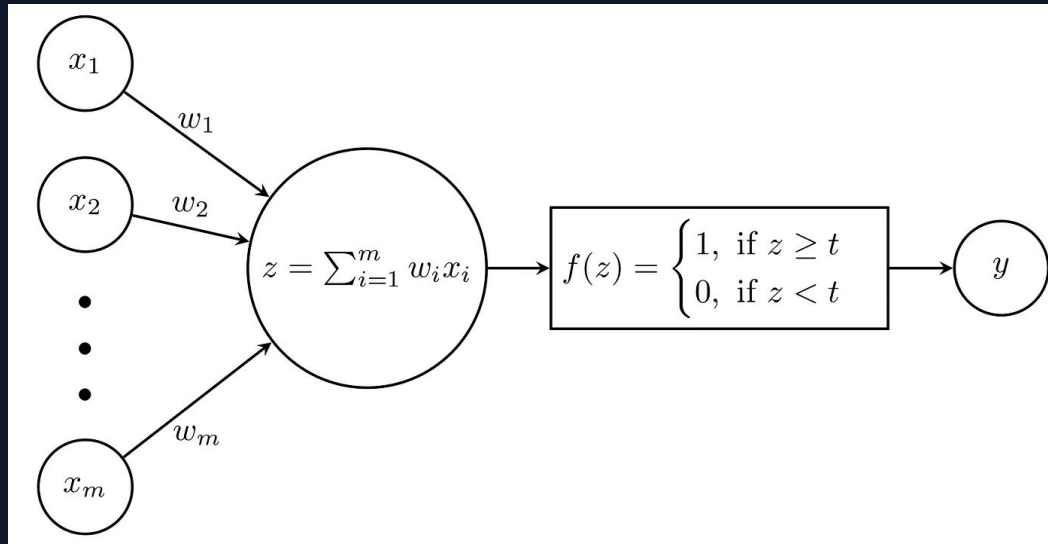


Rosenblatt's improvements to the Perceptron

He added the model's bias, improved weights and developed a learning algorithm (1958-1962).

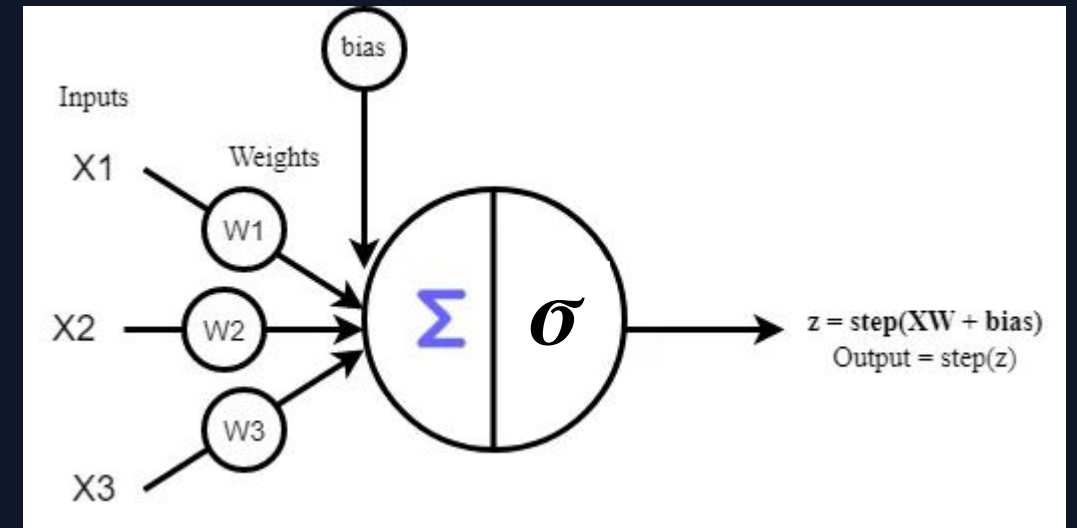
Two Perceptrons

McCulloch-Pitts'



- No bias term
- Only binary outputs
- No learning (parameter adjusting)

Rosenblatt's



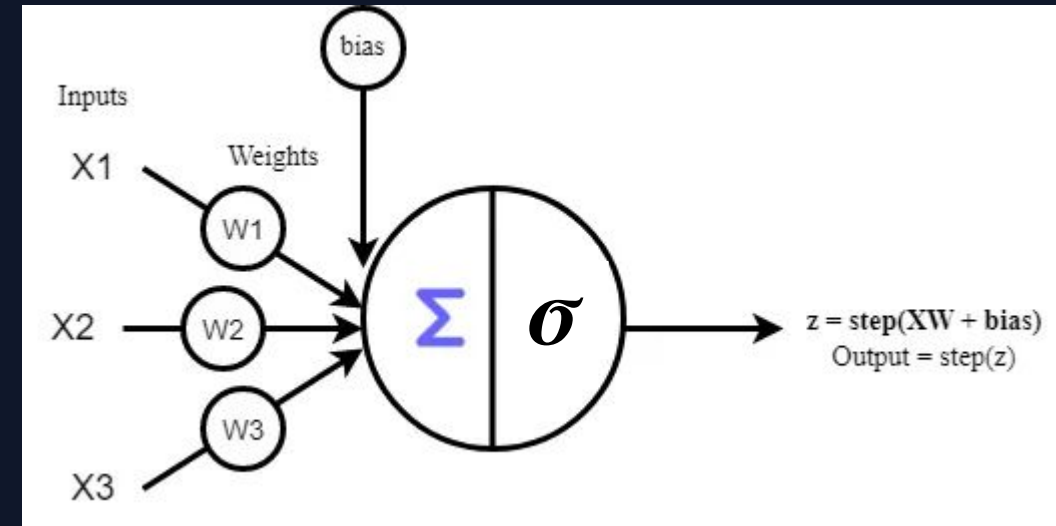
- Bias term added
- Parameters can be updated (!!)
- Outputs can be continuous (activation function σ)

The perceptron maths

Weighing and summation (forward pass)

$$z = \sum_{i=1}^n w_i x_i + b$$

$$z = \mathbf{w}^T \mathbf{x} + b$$



Learning = adjusting parameters

$$w_i \leftarrow w_i + \alpha(y - \hat{y})x_i$$

$$\mathbf{w} \leftarrow \mathbf{w} + \alpha(y - \hat{y})\mathbf{x}$$

$$b \leftarrow b + \alpha(y - \hat{y})$$

Recall

$$\hat{y} = \sigma(z)$$

Why it matters?

- Working with continuous predictions.
- The basic computation unit for more complex architectures = MLP, DNN, CNN, RNN.
- New paradigm = learning as adjustment of weights.

Applications:

1. Predictions (linear or non-linear according to activation function).
2. Classification (Binary).

References

1. Russell, S. and Norvig, P. (2020). Artificial intelligence. A Modern Approach. Pearson.
2. Geron, A. (2019) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. 2nd Edition, O'Reilly Media, Inc.
3. Manndsylva. (2024, August 10th). From theory to practice: Comparing the McCulloch-Pitts and Rosenblatt perceptron models. Medium.
<https://medium.com/@manndsylva/from-theory-to-practice-comparing-the-mcculloch-pitts-and-rosenblatt-perceptron-models-d5c01da507e9>